Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2077

Bachelor of Science in Computer Science and Information Technology

Course Title: **Digital Logic**

Code No: **CSC116** Semester: **1st Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Design a combinatorial circuit that generates 9's complement of a BCD number.

2. Implement the following functions using PLAw (A, B, C, D) = ∑(3,12,13)x (A, B, C, D)= ∑ (7,8,9,10,11,12,13,14,15)y (A, B, C, D) = ∑ (0,2,3,4,5,6,7,8,10,11,15)z (A, B, C, D) = ∑ (1,2,8,12,13)

3. Design sequential circuit specified by the following state diagram using T flip-flops.

Section B

Attempt any Eight questions[8×5=40]

4. List two major characteristics of digital computer. Represent -6 (negative six) using 8 bits in signed magnitude, signed-1's-complement and signed-2's-complement respectively. Represent decimal number 4673 in a) octal, and b) BCD.

5. Where is CMOS suitable to use? Define Power dissipation. Show that the positive logic NAND gate is a negative logic NOR gate and vice versa.

6. Design a full subtractor circuit with three inputs x, y, Binand two outputs Diff and Bout. The circuit subtracts x-y-Binwhere Binis the input borrow, Boutis the output borrow, and Diff is the difference.

7. Design 4-bit even parity generator.

8. What is the difference between a serial and parallel transfer? Explain how to convert serial data to parallel and parallel data to serial. What type of register is needed?

9. Explain negative-edge triggered D flip flop with necessary logic diagram and truth table.

10. Illustrate the use of Binary ripple counter and BCD ripple counter.

11. How binary addition is done? Show binary addition of (11001) with (11011).

12. Write Short notes on (Any two)a) RTLb) State Reductionc) POS